

# Brian Sahagian

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## EDUCATION

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**Boston University, College of Engineering**  
Bachelor of Science in Mechanical Engineering  
GPA: 3.93/4.00 | Dean's List 7/7 Semesters

Boston, MA  
May 2026

## PROJECTS

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**Motor Control Transport Cart**, Electromechanical Design Nov – Dec 2024

- Engineered autonomous cart to carry unsupported vertical aluminum rod 10 feet without tipping
- Integrated 3D printed components, Arduino, and a rotary encoder to optimize speed and stability
- Utilized SolidWorks Motion Analysis to calculate system's maximum stable acceleration profile
- Developed proportional control algorithm to adjust motor output in real time, minimizing velocity error

**Automated Painting Device**, Electromechanical Design Sep – Nov 2024

- Designed and built autonomous 2.5 degree of freedom painting and plotting device for art challenge
- Drafted CAD model of linear stage, integrating Arduino, stepper motors, and 3D printed components
- Developed MATLAB script translating CAM-generated file into precise linear plotting commands
- Recreated any provided image file with selection of 3 paint colors and a sharpie outline onto paper

**Social Network App**, Software Engineering Nov – Dec 2025

- Co-developed C++ desktop application for modeling social networks using weighted attribute similarity
- Designed abstract interfaces to ingest datasets from CSV, JSON, or synthetic sources
- Engineered interactive Qt frontend for visualizing adjacency data and social connections
- Utilized Git for versioning and Google Test for continuous integration and cross-platform development

## SKILLS

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**Programming:** MATLAB, C, C++, Python, SQL, HTML, CSS, JavaScript, ROS/ROS2 Framework

**Design & Simulation:** SolidWorks, SolidWorks Motion, Onshape, COMSOL FEA, Fusion CAM

**Mechanical:** Manual & CNC machining, mill, lathe, 3D printing, precision measurement, material selection

**Technical:** Rapid prototyping, microcontrollers, PID controller, actuators, motion planning, inverse kinematics, control theory, computer vision, stress analysis, algorithms, data structures, Agile, Git, AI integration

## WORK EXPERIENCE

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**Boston University Engineering Product Innovation Center** Boston, MA

Machine Shop Student Advisor Dec 2024 – Present

- Supervise machine shop floor, assist and teach students proper use of machining equipment
- Optimize engineering project designs with machining-conscious and efficiency-driven decisions
- Enforce strict safety protocols to ensure secure shop environment, BU SciShield Lab Safety certified
- Perform routine maintenance and repair of personally assigned Bridgeport manual mill

**Jaros, Baum & Bolles** Boston, MA

HVAC Engineering Intern Jun – Aug 2025

- Drafted and revised HVAC system layouts for 10+ projects according to industry standards
- Verified mechanical installations across 4+ job sites, identifying design-to-field discrepancies
- Conducted cross-functional integration in Revit, resolving 3D model interference across subsystems

## ACTIVITIES

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- Boston University Tau Beta Pi Engineering Honor Society
- Boston University Snowboard Team, competing in slopestyle and slalom events